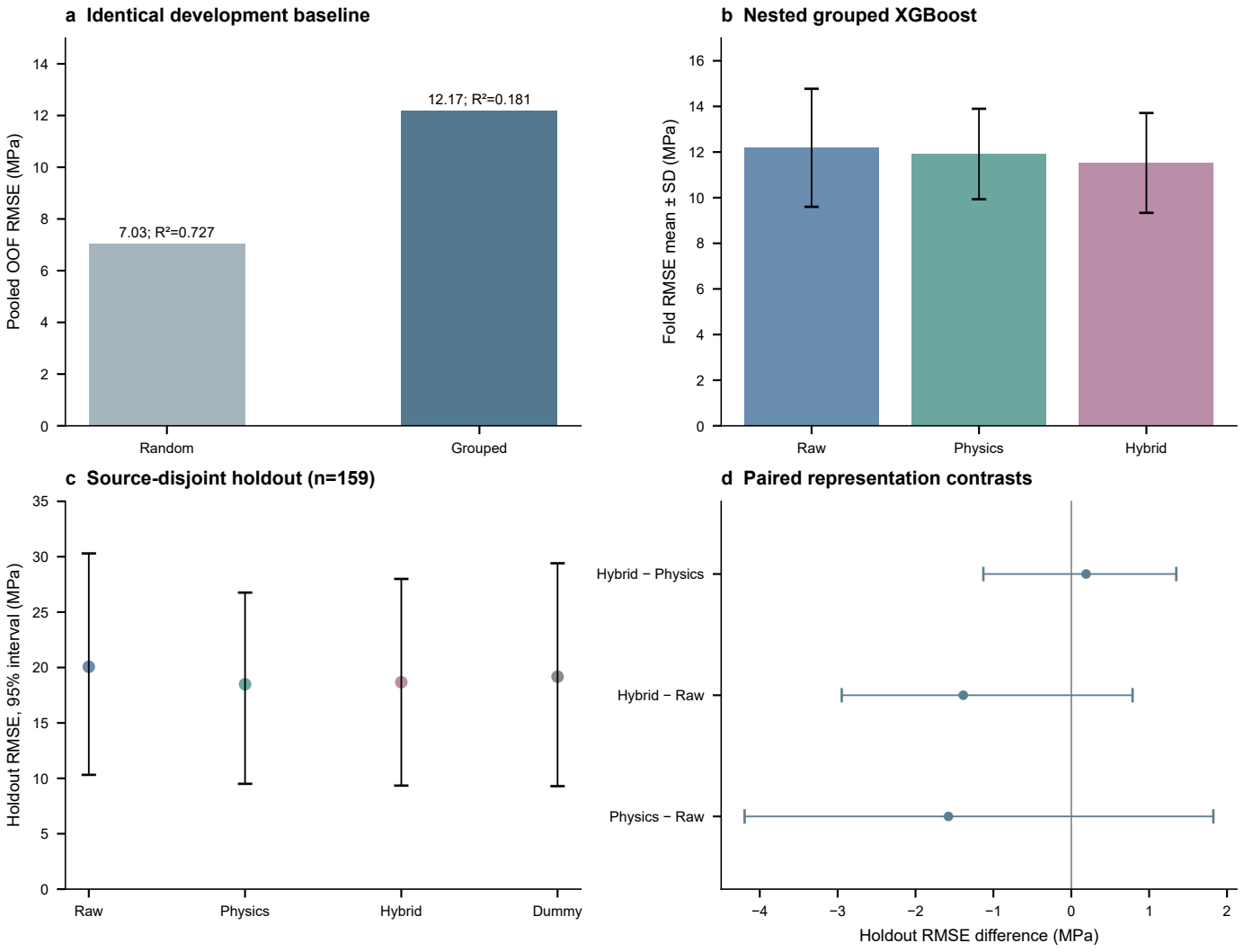

a Identical development baseline
Pooled OOF RMSE (MPa)
7.03; R²=0.727
12.17; R²=0.181
Random
Grouped
b Nested grouped XGBoost
Fold RMSE mean ± SD (MPa)
Raw
Physics
Hybrid
c Source-disjoint holdout (n=159)
Holdout RMSE, 95% interval (MPa)
Raw
Physics
Hybrid
Dummy
d Paired representation contrasts
Hybrid − Physics
Hybrid − Raw
Physics − Raw
Holdout RMSE difference (MPa)